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Project: Residential Power Bill Rates

Company: Alabama Power Company

Website: <https://www.alabamapower.com/content/dam/alabama-power/pdfs-docs/Rates/FD.pdf>

Objectives:

- (1.) Calculate the winter power bill of the residents of Alabama within each range of specific water usage manually using Arithmetic method.
- (2.) Write a piecewise function of the residential rates.
- (3.) Recalculate the same power bill of the residents of Alabama within each range of specific power usage algebraically using the Piecewise Function method.

Information:

Base Charge: \$14.50 per customer; plus

BILLING MONTHS OCTOBER – MAY**Charge for Energy:**

12.4384¢ per kWh for the first 750 kWh, plus

11.2384¢ per kWh for all over 750 kWh.

Conversions:

$$12.4384\text{¢} = \frac{\$12.4384}{100} = \$0.124384 \text{ per } KWh$$

$$11.2384\text{¢} = \frac{\$11.2384}{100} = \$0.112384 \text{ per } KWh$$

Numbers kWh to be tested:

I will test the following utilization of kWh:

(1) 0 kWh

(2) 300 kWh

(3) 1000 kWh

Arithmetic method:

(1.) For the utilization of 0 kWh:

$$Cost = \$14.50$$

(2.) 300 kWh

$$Cost = 14.50 + 0.124384(300)$$

$$Cost = 14.50 + 37.3152$$

$$Cost = 51.8152$$

$$Cost \cong \$51.82$$

(3.) 1000 kWh

$$Cost = 14.50 + 0.124384(750) + 0.112384(1000 - 750)$$

$$\text{Cost} = 14.50 + 93.288 + 0.112384(250)$$

$$\text{Cost} = 14.50 + 93.288 + 28.096$$

$$\text{Cost} = 135.884$$

$$\text{Cost} \cong \$135.88$$

Piecewise Function:

Let us define variables:

Let k be the power consumed in kWh

$e(k)$ is the cost per consumption of power in \$

Base Charge: \$14.50 per customer; plus

First piece: First 750 kWh: $0 \leq k \leq 750$

$$e(k) = 14.50 + 0.124384k$$

$$e(k) = 0.124384k + 14.50$$

Second piece: Over 750 kWh: $0 > k > 750$

$$e(k) = 14.50 + 0.124384(750) + 0.112384(k - 750)$$

$$e(k) = 14.50 + 93.288 + 0.112384k - 84.288$$

$$e(k) = 0.112384k + 23.5$$

Piecewise Function

$$e(k) = \begin{cases} 0.124384k + 14.50; & 0 \leq k \leq 750 \\ 0.112384k + 23.5; & k > 750 \end{cases}$$

Piecewise Function (Algebraic Method):

I will test the same numbers I used with the Arithmetic Method.

(1.) For the utilization of 0 kWh:

0 kWh falls in the first piece

$$e(k) = 0.124384k + 14.50$$

$$e(0) = 0.124384(0) + 14.50$$

$$e(0) = 0 + 14.50$$

$$e(0) = \$14.50$$

(2.) For the utilization of 300 kWh:

300 kWh falls in the second piece

$$e(k) = 0.124384k + 14.50$$

$$e(300) = 0.124384(300) + 14.50$$

$$e(300) = 37.3152 + 14.50$$

$$e(300) = 51.8152$$

$$e(300) \cong \$51.82$$

(3.) For the utilization of 1000 kWh:

1000 kWh falls in the third piece

$$e(k) = 0.112384k + 23.5$$

$$e(1000) = 0.112384(1000) + 23.5$$

$$e(1000) = 112.384 + 23.5$$

$$e(1000) = 135.884$$

$$e(1000) \cong \$135.88$$

Reference (MLA)

Chukwuemeka, Samuel. "Piecewise Function Application: Power Bill." *Appspot.com*, 2018, <https://conferencepresentations.appspot.com/Projects/PreCalculus/PiecewiseFunctions/PowerBill.html>. Accessed 19 Sept. 2024

RATE FD FAMILY DWELLING- RESIDENTIAL SERVICE in Docket # 24860. The KWh Charges Shown Reflect Adjustment pursuant to Rates RSE and CNP for Application to Monthly Bills Effective for June 2023 Billings. PAGE 1 of 2 EFFECTIVE DATE. 2008 from <https://www.alabamapower.com/content/dam/>. Accessed 19 Sept. 2024